

CIG: Measuring Conversational Information Gain in Deliberative Dialogues with Semantic Memory Dynamics

CIG: 在具有语义记忆动态的审议对话中测量会话信息增益

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Abstract

摘要

Measuring the quality of public deliberation requires evaluating not only civility or argument structure, but also the informational progress of a conversation. We introduce a framework for Conversational Information Gain (CIG) that evaluates each utterance in terms of how it advances collective understanding of the target topic. To operationalize CIG, we model an evolving semantic memory of the discussion: the system extracts atomic claims from utterances and incrementally consolidates them into a structured memory state. Using this memory, we score each utterance along three interpretable dimensions: Novelty, Relevance, and Implication Scope. We annotate 80 segments from two moderated deliberative settings (TV debates and community discussions) with these dimensions and show that memory-derived dynamics (e.g., the number of claim updates) correlate more strongly with human-perceived CIG than traditional heuristics such as utterance length or TF-IDF. We develop effective LLM-based CIG predictors paving the way for information-focused conversation quality analysis in dialogues and deliberative success.

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衡量公共讨论的质量不仅需要评估礼貌或论点结构，还需要评估对话中的信息进展。我们引入了一个名为“对话信息增益”（Conversational Information Gain, CIG）的框架，该框架根据每条发言如何促进对目标话题的集体理解来评估每条发言。为了具体化 CIG，我们建模了讨论的演变语义记忆：系统从发言中提取原子主张，并逐步将它们整合为结构化的记忆状态。利用这一记忆，我们从三个可解释的维度对每条发言进行评分：新颖性（Novelty）、相关性（Relevance）和影响范围（Implication Scope）。我们使用这些维度标注了来自两个受调控的讨论场景（电视辩论和社区讨论）中的 80 个片段，并表明由记忆衍生的动力学（例如主张更新的数量）与人类感知的 CIG 相关性比传统启发式方法（如发言长度或 TF-IDF）更强。我们开发了有效的基于大型语言模型（LLM）的 CIG 预测器，为对话和讨论中的信息导向式对话质量分析奠定了基础。¹

1 Introduction

1 引言

Public deliberation—reasoned dialogue aimed at collective understanding and decision-making (Habermas, 1985) for public interest (Dewey and Rogers, 2012)—is fundamental to democratic societies. Yet, the quality of these exchanges, from community forums (Schroeder et al., 2024) to public debates (Montez and Brubaker, 2019), can vary widely from stagnation to productive collaboration.

公共讨论——旨在实现集体理解和决策的理性对话 (Habermas, 1985)，以公共利益为目标 (Dewey 和 Rogers, 2012)——是民主社会的基本要素。然而，从社区论坛 (Schroeder 等, 2024) 到公共辩论 (Montez 和 Brubaker, 2019)，这些交流的质量可以从停滞不前到富有成效的合作差异很大。

Metrics for evaluating dialogue quality (Goddard and Gillespie, 2023) have largely focused on

评价对话质量的指标 (Goddard 和 Gillespie, 2023) 主要集中在

structural formality over substantive content. Approaches based on schemes like the Deliberative Quality Index (DQI; Steenbergen et al. (2003)) or computational proxies for civility (Anuchitanukul et al., 2022; Price et al., 2020) and argument structure (Wachsmuth et al., 2024) struggle to capture true informational progress. These surface-level signals can mislead: civil phrasing can conceal malicious intent (Kruk et al., 2024), and valuable insights might emerge from informal exchanges (Walton, 2008; Bohm and Weinberg, 2004). Such methods fail to distinguish constructive progress from bureaucratic talk (Montez and Brubaker, 2019; Walton, 2003).

结构性形式胜过实质性内容。基于如协商质量指数 (DQI; Steenbergen 等, 2003) 等方案，或基于礼貌的计算代理 (Anuchitanukul 等, 2022; Price 等, 2020) 以及论证结构 (Wachsmuth 等, 2024) 的方法，难以捕捉真正的信息进展。这些表面信号可能会误导：礼貌的措辞可能掩盖恶意意图 (Kruk 等, 2024)，而有价值的见解可能源自非正式交流 (Walton, 2008; Bohm 和 Weinberg, 2004)。此类方法无法区分建设性进展与官僚式谈话 (Montez 和 Brubaker, 2019; Walton, 2003)。

Grounded in Meadow and Yuan’s notion of information impact as “A change, or the nature or magnitude of change, in the knowledge base of a subject domain of the recipient” (Meadow and Yuan, 1997, p. 710), we define Conversational Information Gain (CIG) as the degree to which an utterance advances collective understanding toward the goal/topic. We decompose CIG into three interpretable aspects—Novelty, Relevance, and Implication Scope—capturing whether a contribution introduces new information, connects to the shared goal, and extends its implications to the public community beyond individual cases. To operationalize CIG, we require a representation of the evolving collective knowledge state against which each new utterance can be evaluated: both annotators and models must know what has already been said. As shown in Figure 1, we implement this through a lightweight semantic memory that maintains a consolidated set of claims.

基于 Meadow 和 Yuan 对信息影响的定义，即“接收者所处领域知识库中的变化，或变化的性质和程度” (Meadow and Yuan, 1997, p. 710)，我们将会话信息增益 (Conversational Information Gain, CIG) 定义为话语在推进集体理解向目标/主题方面所达到的程度。我们将 CIG 分解为三个可解释的方面——新颖性 (Novelty)、相关性 (Relevance) 和影响范围 (Implication Scope)，分别捕捉贡献是否引入了新信息、是否与共享目标相关联，以及是否将影响延伸至个体案例之外的公共社区。为了操作化 CIG，我们需要一个能够表示集体知识状态演化的表示方法，以便对每个新的发言进行评估：无论是标注者还是模型，都必须了解之前已经说过什么。如图 1 所示，我们通过一个轻量级的语义记忆来实现这一点，该记忆维护了一组整合的陈述。

We first validate CIG through human annotation of 80 dialogue segments drawn from two moderated group-discussion settings—TV debates and community discussions—achieving moderate-to-high inter-annotator agreement for CIG and its aspects. We then validate automation by using an LLM with access to the same information as the

我们首先通过人工标注的方式验证 CIG，在两个调节过的小组讨论场景——电视辩论和社区讨论中选取了 80 个对话片段，实现了对 CIG 及其各个方面的中等至高水平的标注者间一致性。随后，我们通过使用能够访问与人类相同信息的大型语言模型 (LLM) 来验证自动化方法。

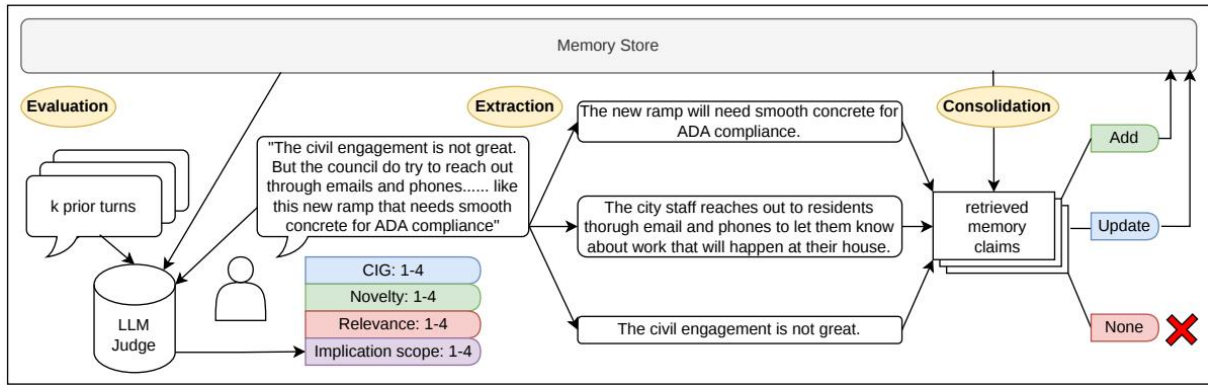


Figure 1: images/7c719315e471b1252d78183d2efe3507909d62add3410b35d7725751184dce78.jpg

Figure 1: Overview of the CIG pipeline. Each utterance is evaluated with the Semantic Memory as knowledge context for Novelty, Relevance, Implication Scope, and overall CIG (1–4). The Semantic Memory is maintained through two modules: Extraction, which converts utterances into atomic claims; and Consolidation, which matches extracted claims against the retrieved memory, triggering ADD, UPDATE, or NONE operations.

图 1: CIG 流程概览。每个语句均以语义记忆作为知识背景，用于评估新颖性、相关性、含义范围以及总体 CIG (1–4)。语义记忆通过两个模块进行维护：提取模块，将语句转换为原子声明；以及巩固模块，将提取的声明与检索到的记忆进行匹配，触发 ADD、UPDATE 或 NONE 操作。

human annotators (topic, short context, and a prior-memory summary), and show that LLM predictions closely track aggregated human judgments. Finally, by varying the model’s prior context, we find that predictions based on retrieved memory summaries are highly correlated with those based on the full preceding transcript, indicating that memory summaries provide a compact yet faithful substitute for full-history context in automated CIG assessment.

人类标注者（主题、简短上下文和先前记忆摘要），并展示出 LLM 的预测结果与聚合的人类判断非常接近。最后，通过改变模型的先前上下文，我们发现基于检索到的记忆摘要的预测与基于完整前序转录文本的预测高度相关，这表明记忆摘要为自动化 CIG 评估提供了紧凑而忠实的完整历史上下文替代方案。

We then analyze how memory dynamics relate to perceived informational progress, finding that simple memory-state signals (e.g., claim update counts) track human CIG ratings more consistently than common heuristics such as sentence entropy or TF-IDF. An unsupervised aggregation analysis reveals a “conjunctive bottleneck”: an utterance’s CIG is effectively limited by its weakest aspect. Finally, we provide a case study illustrating how CIG can be used to analyze downstream interaction dynamics in moderated discussions.

我们随后分析了记忆动态如何与感知到的信息进展相关联，发现简单的记忆状态信号（例如，声明更新次数）比常见的启发式方法（如句子熵或 TF-IDF）更一致地跟踪人类的 CIG 评分。一项无监督的聚合分析揭示了一个“联合瓶颈”：一个陈述的 CIG 实际上受到其最薄弱方面的限制。最后，我们提供了一个案例研究，说明 CIG 可以用于分析受监管讨论中后续互动动态。

2 Related work

2 相关工作

Measuring Informativeness in Conversation While definitions of informativeness vary across disciplines, they share a common conceptual core: the change a message induces in the recipient—whether in certainty (Hunt, 2003), utility (Glazer, 1993), or knowledge state (Meadow and Yuan, 1997). Particularly, dialogue evaluation has largely converged on two necessary conditions re-

quired to trigger this change: novelty (the presence of new signal) and relevance (the alignment of that signal to the goal context) (Ghosal et al., 2022; Maës et al., 2024). However, these two dimensions alone are insufficient to capture the magni-

在对话中衡量信息量性 (Informativeness) 时, 尽管不同学科对信息量性的定义有所不同, 但它们共享一个共同的概念核心: 信息对接收者所产生的影响——无论是对确定性 (Hunt, 2003)、效用 (Glazer, 1993) 还是知识状态 (Meadow and Yuan, 1997) 的影响。特别是, 对话评估在触发这种变化所需的两个必要条件上已基本达成共识: 新颖性 (新信息的存在) 和相关性 (该信息与目标情境的一致性) (Ghosal 等, 2022; Maës 等, 2024)。然而, 仅凭这两个维度还无法充分捕捉到信息量性的全部内涵:

tude or worthiness of a contribution (Huang et al., 2020). While metrics like “impact” or “usefulness” attempt to proxy this third dimension, they are often vaguely defined or rely on subjective Likert scales that conflate personal preference with collective value (Lee et al., 2022; Qian et al., 2025). A complementary line of work extends classical information theory (Shannon, 1948), leveraging surprisal or information-density to trace information flow (Maës et al., 2022; Giulianelli et al., 2021; Tsipidi et al., 2024). However, such measures only partially align with human-perceived salience (Zarcone et al., 2016; Maës et al., 2022), limiting their precision in localizing information exchanges.

贡献的价值或重要性 (Huang 等, 2020)。虽然诸如“影响”或“有用性”等指标试图代理这一第三维度, 但它们往往定义模糊, 或依赖主观的李克特量表, 将个人偏好与集体价值混为一谈 (Lee 等, 2022; Qian 等, 2025)。另一条研究路线扩展了经典信息论 (Shannon, 1948), 利用惊讶度或信息密度来追踪信息流动 (Maës 等, 2022; Giulianelli 等, 2021; Tsipidi 等, 2024)。然而, 这些度量标准与人类感知的显著性部分一致 (Zarcone 等, 2016; Maës 等, 2022), 限制了它们在定位信息交流方面的精确度。

Agent Semantic Memory Agent memory modules provide a way to measure knowledge acquisition during conversation (Zhang et al., 2024). Originally developed to sustain coherence and personalization in conversational agents over long conversation sessions, these modules function by maintaining a selective, persistent state. In frameworks such as Mem0 (Chhikara et al., 2025), salient claims are extracted from utterances and an LLM applies update operations to a knowledge store. This approach improves temporal and multi-hop reasoning on long-dialogue benchmarks (Maharana et al., 2024) and reduces latency and token cost relative to full-history baselines.

代理语义记忆代理记忆模块提供了一种在对话中衡量知识获取的方法 (Zhang 等, 2024)。最初开发用于在长时间对话会话中维持对话代理的连贯性和个性化, 这些模块通过维护一种选择性的、持久的状态来实现其功能。在诸如 Mem0 (Chhikara 等, 2025) 等框架中, 从话语中提取出显著的声明, 并且大型语言模型 (LLM) 会对知识存储执行更新操作。这种方法在长对话基准测试中提高了时间性和多跳推理能力 (Maharana 等, 2024), 并且相较于全历史基线, 减少了延迟和令牌成本。

3 Conversational Information Gain (CIG)

3 对话信息增益 (CIG)

Following previous studies in group discussion analysis (Dowell et al., 2019), we define Conversational Information Gain (CIG) as how much a response advances the group’s shared understanding of the topic or progress toward the goal, given both

根据之前关于小组讨论分析的研究 (Dowell 等, 2019), 我们将对话信息增益 (Conversational Information Gain, CIG) 定义为: 在考虑双方的情况下, 一个回应多大程度上推进了小组对主题的共同理解或向目标的进展。

Aspect	Label	Definition / Anchor
Conversational Information Gain	1. No gain	Repeats or obstructs; no meaningful advance beyond the existing knowledge.

2. Minimal gain	Small clarification or slight nuance that is noticeable but limited.	
3. Incremental	Adds new details/mechanisms within the same conceptual frame or ideas within the topic.	
4. Insightful	Reframes or introduce new ideas under the topic; shifts the conversation in a new valuable way.	
Novelty	1. Not novel	Repetition/paraphrase of prior content or non/common-sense content.
2. Minimally novel	Minor or mostly predictable detail added to an existing idea.	
3. Moderately novel	New evidence, concrete example, or supporting detail expanding an existing idea.	
4. Highly novel	New framework, principle, idea, or line of reasoning that opens a new direction.	
Relevance	1. Not relevant	Off-topic; no connection to the conversation goal.
2. Minimally relevant	Loose or indirect link; requires inference to connect.	
3. Moderately relevant	Substantially related but not central (e.g., side issue or counterpoint).	
4. Highly relevant	Directly and explicitly addresses the core topic or goal.	
Implication Scope	1. Local	Manages the immediate moment; implication limited to participants/procedures.
2. Bounded	Self-contained fact, feeling, or stance; no generalization beyond the case.	
3. Generalizing	Inductively generalizes a case or evidence to a broader audience.	
4. Universal	States an abstract principle, value, or norm with wide or universal applicability.	

方面	标签	定义 / 锚点
对话信息增益	1. 没有收益	重复或阻碍；没有超出现有
2. 最小收益	小的澄清或细微的差别，虽然可以察觉但范围有限。	

3. 增量式	在相同的概念框架或主题内的新细节/机制，或在同一主题内的新想法。	
4. 有洞察力的新颖性	在主题下重新表述或引入新观点；以有价值的新方式转变对话。	重复/改写先前内容或非常识
1. 最小化创新	1. 不新颖	
2. 最小化创新	对现有想法添加的小细节或几乎可以预测的细节。	
3. 中等程度的创新	新证据、具体例子或支持性细节，用于扩展已有观点。	
4. 高度创新	新开辟方向的新框架、原则、理念或推理思路。	
相关性	1. 无关	离题；与对话目标无关。
2. 最小相关	松散或间接的联系；需要进行推断才能建立联系。	
3. 中度相关	实质性相关但非核心（例如：次要问题或相反观点）。	
4. 高度相关	直接且明确地针对核心主题或目标。	
影响范围	1. 本地	管理即时时刻；影响仅限于
2. 有界	一个独立的事实、感受或立场；不超出具体案例进行一般化。	
3. 推广至一般情况	从一个案例或证据中归纳出更广泛的结论。	
4. 通用性	陈述一个具有广泛或普遍适用性的抽象原则、价值或规范。	

Table 1: Conversational Information Gain (CIG) rubric. Each aspect is rated on a 1–4 scale using Prior Knowledge and the preceding dialogue as the collective knowledge context.

表 1：对话信息增益（CIG）评分标准。每个方面都使用先验知识和之前的对话作为集体知识背景，采用 1–4 的评分标准进行评分。

prior knowledge and the preceding dialogue. We decompose CIG into three aspects—Novelty, Relevance, and Implication Scope—to connect general criteria for informativeness to the demands of public deliberation. Novelty and Relevance capture whether a contribution introduces new information and is tied to the discussion goal, while Implication Scope captures how broadly the contribution’s implications extend beyond the immediate case and people, reflecting the public orientation of deliberation. All constructs are rated on a four-level scale (Table 1). Importantly, CIG reflects informational advancement rather than conversational quality per se—reiterations or coordination moves typically score low on CIG, even though they may still support the discussion indirectly.

先前的知识和对话历史。我们将 CIG 分解为三个方面——新颖性、相关性和影响范围——以将一般性的信息性标准与公众讨论的需求联系起来。新颖性和相关性捕捉了贡献是否引入了新的信息以及是否与讨论目标相关联，而影响范围则捕捉了贡献的影响在多大程度上超越了当前案例和人们本身，反映了讨论的公众导向。所有构念都使用四等级量表进行评分（表 1）。重要的是，CIG 反映的是信息进展而非对话质量本身——重复或协调性行为通常在 CIG 中得分较低，即使它们可能仍然间接支持讨论。

To anchor high CIG levels, we draw on Chi (2009)’s typology of conceptual change which distinguishes assimilation—adding information within an existing mental model—from accommodation—restructuring the model itself. Accordingly, we separate our top two CIG levels: Level 3 (Incremental) reflects assimilation, where an utterance adds evidence, details, or mechanisms within the current framing; Level 4 (Insightful) reflects accommodation, where an utterance introduces a new framing or principle that qualitatively redirects the discussion. Detailed definitions for each aspect are in Table 1, and examples in Appendix Table 12.

为了锚定高水平的 CIG（概念性变化指标），我们借鉴 Chi（2009）对概念性变化的分类，该分类区分了同化（assimilation）与顺应（accommodation）。同化是指在现有的心理模型内添加信息，而顺应则是对模型本身进行重构。据此，我们将 CIG 的前两个最高水平分开：Level 3（渐进式）反映同化，其中话语在当前框架内添加证据、细节或机制；Level 4（洞察式）反映顺应，其中话语引入新的框架或原则，从而在质上改变讨论的方向。每个方面的详细定义见表 1，示例见附录表 12。

Novelty Assesses whether the information is new compared to the prior knowledge and preceding dialogue. Traditional approaches often proxy novelty with n-gram overlap against prior text (Soboroff and Harman, 2005), but recent work shows that

新颖性评估信息与先前知识和对话相比是否新颖。传统方法通常通过与先前文本的 n-gram 重叠来代理新颖性 (Soboroff 和 Harman, 2005)，但近期的研究表明，

lexical novelty correlates poorly with human perception (Saakyan et al., 2025). Consequently, we rate novelty on a similar four-level scale as CIG, with levels reflecting the message's effect on the existing conceptual framing. The novelty score is intentionally independent of topical alignment and magnitude of impact.

词汇新颖性与人类感知的相关性较差 (Saakyan 等, 2025)。因此，我们采用与 CIG 相似的四级尺度来评估新颖性，各级反映信息对现有概念框架的影响。新颖性评分有意与主题一致性及影响程度无关。

Relevance Measures how substantively a message relates to the main conversation topic or goal. Since lexical-overlap proxies correlate weakly with human judgments in context-sensitive and implicit dialogue (Yeh et al., 2021; Dascal, 1977), we adopt the four-level topic-conversation relevance scale (Fan et al., 2024). Levels 1 (off-topic) and 4 (directly on-topic) form the endpoints; the key distinction is between the middle tiers. Level 2 (minimally relevant) covers content whose connection to the goal is indirect and requires a bridging inference (e.g., “school choice” for topic “housing affordability”), and Level 3 covers content that is clearly connected but not central, typically a recognized subtopic (e.g., “zoning restrictions”).

相关性衡量信息内容与主要对话主题或目标之间的实质性关联程度。由于词汇重叠度代理在情境敏感和隐含对话中与人类判断的相关性较弱 (Yeh 等, 2021; Dascal, 1977)，我们采用四级主题—对话相关性量表 (Fan 等, 2024)。其中，级别 1 (离题) 和级别 4 (直接相关) 构成两端点；关键区别在于中间层级。级别 2 (最小相关) 涵盖与目标间接相关、需要进行桥梁式推理的内容 (例如，主题为“住房可负担性”时，“择校”)，级别 3 涵盖明显相关但并非核心内容，通常是被认可的子主题 (例如，“分区限制”)。

Implication Scope Measures the intended reach of a statement—who it is meant to matter to and how far its implications generalize. Many dialogue-evaluation schemes include an “impact” dimension (Lee et al., 2022). In public deliberation, however, perceived impact is subjective and highly context-dependent. We therefore operationalize this idea—drawing on the notion of “generality” in Meadow and Yuan (1997)—as the scope of implication a contribution projects beyond the immediate speaker and case. This is deliberation-motivated:

含义范围衡量陈述的预期影响范围—它旨在影响谁，以及其含义在多大程度上具有普遍性。许多对话评估方案都包含一个“影响”维度 (Lee 等, 2022)。然而，在公共讨论中，感知到的影响是主观的，并且高度依赖于具体情境。因此，我们借鉴 Meadow 和 Yuan (1997) 提出的“普遍性”概念，将这一理念操作化为贡献所投射的含义范围，即超越说话者本人和具体案例的范围。这种操作化是基于讨论动机的：

participants often move between situated testimony (what happened to me/us here) and public reasons (what should matter to the community), and this “lifting” from particulars to publicly addressable concerns is commonly treated as central to deliberative legitimacy (Habermas, 1985).

参与者经常在具体证言 (对我/我们在这里发生了什么) 和公共理由 (什么应该引起社区关注) 之间转换，这种从具体细节提升到可以公开讨论的问题的过程，通常被视为协商民主合法性 (Habermas, 1985) 的核心。

Our rubric has four levels. Level 1 covers local or procedural moves whose significance is confined to the immediate interaction (e.g., turn management). Level 2 captures bounded, case-specific content (e.g., a personal fact or stance) that does not generalize beyond the case. Level 3 reflects inductive generalization, where a specific experience or piece of evidence is used to motivate a broader pattern—such as testimony offered to evidence social harm (Kessler et al., 2023). Level 4 denotes universal, principle-level claims framed for wide public applicability (e.g., rights, fairness, norms). We emphasize that this hierarchy is descriptive rather than normative: effective deliberation often requires a rhythm between grounded particulars (Levels 2–3) and abstract principles (Level 4). Accordingly, higher scope does not necessarily imply greater infor-

mativeness; rather, Scope characterizes how a contribution frames its relevance to the public, which we later test as a potential (but not guaranteed) driver of perceived CIG.

我们的评估标准分为四个层次。第一层次涵盖局部或程序性的行为，其意义局限于即时的互动（例如，回合管理）。第二层次捕捉到有限的、具体案例的内容（例如，一个个人事实或立场），这些内容无法推广到其他案例之外。第三层次反映了归纳性概括，即通过一个具体经历或证据来推动更广泛模式的形成——例如，通过证词来证明社会危害（Kessler 等，2023）。第四层次表示普遍性的、原则层面的主张，这些主张旨在具有广泛的公共适用性（例如，权利、公平、规范）。我们强调，这种层次结构是描述性的而非规范性的：有效的讨论往往需要在具体细节（第 2-3 层次）和抽象原则（第 4 层次）之间保持一种节奏。因此，更高的范围并不一定意味着更大的信息量；相反，范围描述了贡献如何将其与公众相关联，我们随后将这一方面作为潜在（但不保证）影响感知 CIG 的驱动因素进行检验。

4 Entailment-based Semantic Memory

4 基于蕴含的语义记忆

To automatically and incrementally score the CIG of conversational utterances, we require a dynamic representation of what the conversation has already established. To this end, we extend the Mem0 framework (Chhikara et al., 2025) to a multi-party setting, maintaining an evolving semantic memory that tracks how each speaker introduces and revises claims over time. As shown in Figure 1, the memory is managed through a two-stage process of claim extraction and consolidation to provide the knowledge context for evaluating Novelty, Relevance, Implication Scope, and overall CIG.

为了自动且逐步地对对话中的话语的 CIG（对话信息量）进行评分，我们需要一种动态表示，以反映对话已经建立的内容。为此，我们将 Mem0 框架（Chhikara 等，2025）扩展到多方设置中，维护一个不断演进的语义记忆，以跟踪每位说话者随时间推移如何提出和修改主张。如图 1 所示，该记忆通过一个两阶段过程进行管理，即主张提取和整合，以提供评估新颖性、相关性、含义范围以及整体 CIG 的知识背景。

Claim Extraction The first stage uses an LLM to decompose each utterance into a set of discrete, self-contained claims, converting context-dependent dialogue into a structured representation for semantic memory consolidation and storage (Appendix Table 13 details the prompt). To make each claim interpretable in isolation, the extractor uses local context to resolve ambiguities (e.g., coreference) and splits compound statements into atomic propositions, while normalizing away conversational fillers and hedging to recover the core propositional content (e.g., “I feel that maybe the transport cost is too high” $\rightarrow$ “The transport cost is too high”). Although this normalization removes epistemic markers, it enables more reliable NLI-based entailment checks; we preserve speaker-specific beliefs by storing each extracted claim together with its source speaker and inferred addressee in the memory state.

声明提取第一阶段使用一个大语言模型（LLM），将每个对话内容分解为一组独立、自包含的声明，将依赖上下文的对话转换为结构化的表示形式，用于语义记忆的巩固和存储（附录表 13 详细说明了提示语）。为了使每个声明在孤立情况下都能被解释，提取器使用局部上下文来解决歧义（例如，指代关系），并将复合语句分解为原子命题，同时去除会话中的填充词和缓和语，以恢复核心命题内容（例如，“我感觉运输费用可能太高了” $\rightarrow$ “运输费用太高了”）。尽管这种规范化去除了知识性标记，但它使得基于自然语言推理（NLI）的蕴含检查更加可靠；我们通过在记忆状态中存储每个提取的声明及其来源说话人和推断的接收者，来保留说话人特定的信念。

Multi-Party Memory Consolidation To manage redundancy and track knowledge evolution, each newly extracted claim A is compared against the semantic memory by first retrieving the top-k most semantically similar candidates $\{B_i\}_{i=1}^k$ via vector search. An LLM-based NLI judge bidirectionally evaluates the claim and each candidate, mapping each pair to one of five relations: equivalent, forward_entail, backward_entail, contradiction, or neutral.

多方记忆巩固为了管理冗余并跟踪知识的演进，每个新提取的声明 A 都会首先通过向量搜索检索出最语义相似的前 k 个候选对象 $\{B_i\}_{i=1}^k$ ，然后与语义记忆进行比较。基于大型语言模型（LLM）的自然语言推理（NLI）裁判从两个方向评估该声明和每个候选对象，将每对映射到五种关系中的一种：等价、前向蕴

含、后向蕴含、矛盾或中性。

We then apply a deterministic policy to select an action $\alpha(A) \in \{\text{ADD}, \text{UPDATE}, \text{NONE}\}$. To strictly track speaker-specific belief trajectories without collapsing inter-speaker disagreements, we restrict operations to the same-speaker subset $S(A) = \{B_i \mid \text{speaker}(B_i) = \text{speaker}(A)\}$. If $S(A) = \emptyset$ or contains only neutral relations, we trigger ADD and insert A into the semantic memory; this ensures that parallel framings or contested claims from other participants are preserved as distinct entries. For $S(A) \neq \emptyset$, equivalence or backward entailment triggers NONE (taking priority over updates), while contradiction or forward entailment triggers UPDATE on the single most similar item $B^* = \arg \max_{B \in S(A)} \text{sim}(A, B)$. Here, UPDATE means revising the existing memory B^* using the new claim A : contradictions replace B^* with A , whereas forward entailments merge A and B^* into a more specific consolidated claim. Appendix Table 11 summarizes this mapping with examples, and Appendix Table 14 details the prompt.

我们随后应用一个确定性策略来选择一个动作 $\alpha(A) \in \{\text{ADD}, \text{UPDATE}, \text{NONE}\}$ 。为了严格跟踪特定说话人的信念轨迹，而不使不同说话人之间的分歧合并，我们将操作限制在同一个人的子集 $S(A) = \{B_i \mid \text{speaker}(B_i) = \text{speaker}(A)\}$ 。如果 $S(A) = \emptyset$ 或包含仅中性关系，我们将触发 ADD 并将 A 插入到语义记忆中；这确保了其他参与者的平行表述或争议性主张被保留为独立的条目。对于 $S(A) \neq \emptyset$ ，等价或逆蕴含会触发 NONE（优先于更新），而矛盾或正向蕴含则会触发对最相似的单一项目 $B^* = \arg \max_{B \in S(A)} \text{sim}(A, B)$ 的更新。在这里，UPDATE 表示使用新的主张 A 来修订现有的记忆 B^* ：矛盾会将 B^* 替换为 A ，而正向蕴含则会将 A 和 B^* 合并为一个更具体的综合主张。附录表 11 用示例总结了这种映射关系，附录表 14 详细说明了提示内容。

5 Data and Annotation

5 数据与标注

We annotated transcripts from two deliberative discussion settings: ten TV debates from INTELLIGENCE SQUARED (INSQ; Zhang et al. (2016)) and ten community discussions from the FORA corpus (FORA; Schroeder et al. (2024)) about Durham’s community future vision. To mitigate annotator bias and potential harms, we manually curated this set by excluding topics of a highly sen-

我们对两种协商讨论情境下的转录文本进行了标注：来自 INTELLIGENCE SQUARED (INSQ; Zhang 等, 2016) 的十场电视辩论，以及来自 FORA 语料库 (FORA; Schroeder 等, 2024) 的十场社区讨论，这些讨论涉及杜伦社区未来愿景。为减轻标注者偏见和潜在危害，我们通过手动筛选，排除了高度敏感的话题。

	FORA	INSQ
Session-level (10 per setting, avg $\pm$ std)		
Utterances / Session	210.1 $\pm$ 56.5	205.8 $\pm$ 73.8
Words / Utterance	38.8 $\pm$ 11.9	36.4 $\pm$ 17.9
Speakers / Session	6.8 $\pm$ 1.0	8.8 $\pm$ 1.8
Annotated segments (4 per setting, avg $\pm$ std)		
Utterances / Segment	11.6 $\pm$ 4.4	11.3 $\pm$ 4.7
Speakers / Segment	4.5 $\pm$ 1.3	4.4 $\pm$ 1.0
Words / Non-skipped Utterance	86.3 $\pm$ 52.5	88.6 $\pm$ 56.0
Speaker Gini	0.42 $\pm$ 0.16	0.21 $\pm$ 0.15
Skipped Tokens (%)	17.8 $\pm$ 21.0	9.4 $\pm$ 9.0

	FORA	INSQ
会话级别（每设置 10 个，平均 $\pm$ 标准差）		
发言 / 会话	210.1 $\pm$ 56.5	205.8 $\pm$ 73.8
词语 / 话语	38.8 $\pm$ 11.9	36.4 $\pm$ 17.9
演讲者 / 会场	6.8 $\pm$ 1.0	8.8 $\pm$ 1.8
注释片段（每种设置 4 个，平均值 $\pm$ 标准差）		
发言 / 段落	11.6 $\pm$ 4.4	11.3 $\pm$ 4.7
发言者 / 段落	4.5 $\pm$ 1.3	4.4 $\pm$ 1.0
词语 / 未跳过的语句	86.3 $\pm$ 52.5	88.6 $\pm$ 56.0
演讲者基尼	0.42 $\pm$ 0.16	0.21 $\pm$ 0.15
跳过的标记 (%)	17.8 $\pm$ 21.0	9.4 $\pm$ 9.0

Table 2: Session- and segment-level descriptive statistics for the annotated FORA and INSQ dialogues. For segment-level statistics, moderator turns and short fragmentary utterances marked as skipped are excluded when computing Words / Non-skipped Utterance and Speaker Gini. The speaker-participation Gini coefficient measures inequality in contribution within a segment, with higher values indicating more uneven participation.

表 2：标注的 FORA 和 INSQ 对话的会话级和片段级描述性统计信息。对于片段级统计信息，在计算 Words / 非跳过发言和说话人基尼指数时，排除了调停者发言和被标记为跳过的简短断续发言。说话人参与度基尼系数衡量了片段内贡献的不平等程度，数值越高表示参与度越不均衡。

sitive or recently polarizing nature. While both settings feature a moderator and multi-party discussion (see Table 2 for descriptive statistics), they differ in atmosphere and intent. INSQ debates are competitive events organized around a motion and two teams, featuring domain experts in which the moderator primarily coordinates turn-taking. FORA assemblies are collaborative, brainstorming meetings where facilitators often participate as peers alongside general community members. This contrast is also visible in the speaker Gini values in Table 2: INSQ shows more even participation, consistent with tighter moderation, whereas FORA is more unequal, reflecting a freer and more community-driven discussion style.

具有敏感性或近期引发争议的性质。虽然两种设置都包含主持人和多方讨论（参见表 2 中的描述性统计），但它们在氛围和意图上有所不同。INSQ 辩论是竞争性活动，围绕一个议题和两支队伍进行，邀请领域专家参与，主持人主要负责协调发言顺序。FORA 会议则是协作性的头脑风暴会议，主持人通常以与普通社区成员同等的身份参与。这种差异在表 2 中的发言者 Gini 值中也可见：INSQ 显示出更均衡的参与度，这与更严格的主持相一致，而 FORA 则更加不均衡，反映了更自由、更具社区驱动性的讨论风格。

5.1 Preprocessing

5.1 预处理

Both FORA and INSQ exhibit a comparable three-part format—introduction, discussion, and conclusion. For INSQ, we use the dataset’s corresponding predefined structure and analyze only the discussion phase. In FORA the boundaries are not annotated, so we use GPT-5 (OpenAI, 2025) to detect them and manually verify. In both settings, conclusions are discarded, and introductions are used to initialize the initial knowledge in the memory modules. We then segment the discussion phase of each episode into sub-topical units using GPT-5 with a shifting window and majority vote (see Appendix A.1 for segmentation details).

FORA 和 INSQ 都表现出类似的三部分格式——引言、讨论和结论。对于 INSQ，我们使用数据集对应的预定义结构，并仅分析讨论阶段。在 FORA 中，边界未被标注，因此我们使用 GPT-5 (OpenAI, 2025)

来检测这些边界，并手动验证。在两种情况下，结论都被丢弃，引言被用来初始化记忆模块中的初始知识。然后，我们使用 GPT-5 通过滑动窗口和多数投票的方法，将每个情节的讨论阶段分割为子主题单元（有关分割的详细信息，请参见附录 A.1）。

For each segment, we generate a prior memory summary by retrieving relevant existing memories via semantic similarity and summarizing them using GPT-5. We developed the summarization prompt using pilot data annotated by one paper author (see Appendix A.3 for development details). To focus the annotation on participants’ contributions, moderator turns and short utterances (< 5 words) (e.g. “I want to-.”) are marked as skipped, accounting for 9–18% of tokens (see Table 2). From each episode, we select four segments for annotation, balancing factors like estimated reading time, skip ratio, and segmentation confidence (see Appendix A.2 for the selection criteria).

对于每个片段，我们通过语义相似性检索相关的现有记忆，并使用 GPT-5 对其进行总结，生成一个先验记忆摘要。我们使用一位论文作者标注的试点数据开发了摘要提示（详见附录 A.3 中的开发细节）。为了聚焦于参与者的贡献，调停者的发言和简短的句子（< 5 个词）（例如 “I want to-.”）被标记为跳过，占 9–18% 的标记（详见表 2）。从每个情节中，我们选择四个片段进行标注，平衡诸如估计阅读时间、跳过比例和分段置信度等因素（详见附录 A.2 中的选取标准）。

	CIG	Novelty	Relevance	Scope
INSQ	0.589	0.506	0.669	0.597
FORA	0.567	0.583	0.566	0.510

	CIG	新颖性	相关性	范围
INSQ	0.589	0.506	0.669	0.597
FORA	0.567	0.583	0.566	0.510

Table 3: Krippendorff’s α by corpus for CIG (Informativeness) and its three aspects.

表 3: CIG（信息性）及其三个方面的 Krippendorff’s α 按语料库划分。

5.2 Annotation Protocol

5.2 注释协议

We recruited 88 annotators on Prolific (£10.60/hour). All participants were required to be native English speakers with at least a bachelor’s degree. We ultimately dropped 10 participants due to consistently low agreement with peers.² Each assignment (approximately 40 minutes) included a tutorial and a prescreen test, followed by four sampled segments from the same session presented in chronological order.

我们在 Prolific 上招募了 88 名标注者（时薪 £10.60）。所有参与者都必须是英语母语者，并且至少拥有学士学位。由于与同行的一致性始终较低，我们最终剔除了 10 名参与者。² 每个任务（大约 40 分钟）包括一个教程和一个预筛选测试，随后是按时间顺序呈现的来自同一会话的四个采样片段。

We deployed two annotation variants over the same material: one collected a single score for overall CIG per utterance (on a scale of 1–4); the other collected three separate ratings for Novelty, Relevance, and Implication Scope, each on a scale of 1–4. The annotation interface showed the topic at the top, with long-term memory (the prior memories summary) and short-term context (three preceding utterances) in a left panel, and the current target utterance for rating in a right panel (see Appendix A.5 for the interface snippet and design). Inter-annotator agreement,

reported in Table 3, is moderate to high across all aspects and settings. Appendix Figure 5 shows the normalized label distributions.

我们在同一材料上部署了两种标注变体：一种为每个语句收集一个总体 CIG 的单一评分（评分范围为 1–4）；另一种则收集三个独立评分，分别对应新颖性（Novelty）、相关性（Relevance）和含义范围（Implication Scope），每个评分范围也为 1–4。标注界面顶部显示了主题，左侧面板显示了长期记忆（即先前的记忆摘要）和短期上下文（前三句话），右侧面板显示了当前需要评分的目标语句（参见附录 A.5 中的界面片段和设计）。表 3 中报告的标注者间一致性，在所有方面和设置中均中等到较高。附录图 5 展示了标准化的标签分布。

6 Validating the CIG Aspects

6 验证 CIG 方面

To test whether overall CIG is explained by the three proposed aspects, we fit ordinal regression models on the 80 annotated segments and compare different predictor sets against a word-count baseline (Table 4). AIC-based model selection suggests that Novelty and Relevance account for most of the predictive signal. In contrast, Implication Scope—which we included to capture a deliberation-specific intuition that contributions with broader public reach might be perceived as more informative—provides little reliable benefit. In INSQ, the best fit is achieved by removing Scope, indicating that including it slightly degrades predictive power; in FORA, the full model improves only marginally over the Scope ablation.

为了检验整体 CIG 是否由提出的三个方面所解释，我们在 80 个标注的片段上拟合有序回归模型，并将不同的预测变量集与词数基线进行比较（表 4）。基于 AIC 的模型选择表明，新颖性和相关性解释了大部分预测信号。相比之下，我们加入的“含义范围”（Implication Scope）旨在捕捉一种特定的思考直觉，即具有更广泛公众影响的贡献可能被认为更有信息量，但其提供的可靠益处却很小。在 INSQ 中，通过移除“含义范围”可以达到最佳拟合效果，这表明包含它会略微降低预测能力；而在 FORA 中，完整模型仅在移除“含义范围”后略有改进。

Novelty and Relevance behave as expected and explain most of the variance in perceived CIG. In contrast, Implication Scope contributes little. This challenges our deliberation-motivated hypothesis that statements framed with broader public reach would systematically be judged as more informative than local or case-specific remarks. One plausible explanation is that informativeness is sensitive to grounding and evidential value, not only generality: low-scope utterances can be highly informative when they introduce concrete facts or lived experiences that audiences treat as diagnostic, whereas high-scope utterances can be uninformative when they restate abstract values without new content. This interpretation aligns with findings that hearing personal experiences can improve social evaluations compared to hearing personal opinions, particularly under polarization, suggesting that situated testimony can carry distinctive communicative value even without being framed as universal public principle (Kessler et al., 2023). Accordingly, Scope may be better interpreted as a descriptive axis of deliberative style rather than a monotonic driver of perceived CIG.

新颖性和相关性表现如预期，能够解释大多数关于感知 CIG（可能指“公民参与”或“公民意见”等概念）的方差。相比之下，影响范围的贡献较小。这挑战了我们的基于反思的假设，即那些具有更广泛公众传播范围的陈述会被系统性地判断为比局部或具体案例的陈述更具信息性。一个合理的解释是，信息性不仅取决于普遍性，还取决于其基础性和证据价值：当陈述引入观众视为诊断性（具有诊断意义）的具体事实或亲身经历时，即使其范围较小，也可能具有高度信息性；而当陈述重复抽象的价值观而没有新的内容时，即使其范围较大，也可能缺乏信息性。这种解释与以下发现相一致：听到个人经历可以提高社会评价，相较于听到个人观点，尤其是在观点极化的情况下，这表明即使不被表述为普遍的公共原则，情境性的证词也可能具有独特的沟通价值（Kessler 等，2023）。因此，范围可能更适合作为一种描述性轴线，用于解释反思性风格，而不是作为感知 CIG 的单调驱动因素。

7 Prediction of CIG and Heuristic Aggregation

7 CIG 预测与启发式聚合

Predicting CIG with GPT-5 We next test whether segment-level CIG annotation can be automated with GPT-5 under the same information conditions as our human annotators. For each target segment, GPT-5 receives (i) the memory-based prior summary, (ii) the three preceding utterances, and (iii) the tutorial exemplars presented to annotators as few-shot demonstrations, and outputs ratings (Table 1) for overall CIG and its aspects (Appendix Tables 15 and 16 detail the prompts). To contextualize performance, we report a human leave-one-out (LOO) baseline, computed as each annotator’s MAE against the mean of the remaining annotators and averaged across annotators, which estimates the expected deviation of an individual annotator from group consensus. As shown in Table 5, GPT-5’s MAE under this matched-context setup is comparable to—and in several cases lower than—human LOO error, suggesting that GPT-5 can reproduce aggregate human judgments at least as well as a typical annotator when constrained to the same prior context (additional models are reported in Appendix Table 18).

使用 GPT-5 预测 CIG 我们接下来测试在与我们的标注员相同的信息条件下，是否可以利用 GPT-5 自动完成段级 CIG 标注。对于每个目标段落，GPT-5 接收 (i) 基于记忆的先验摘要，(ii) 前三个对话内容，以及 (iii) 提供给标注员作为少量示例的教程范例，并输出对整体 CIG 及其各个方面的评分（表 1）。附录中的表 15 和 16 详细列出了提示语。为了对性能进行定位，我们报告了一个人类留一法（LOO）基线，该基线计算为每个标注员的平均绝对误差（MAE）相对于其余标注员的平均值，并在标注员之间进行平均，这估计了单个标注员与群体共识之间的预期偏差。如表 5 所示，在这种匹配上下文设置下，GPT-5 的 MAE 与人类 LOO 错误相当，并在某些情况下更低，这表明当 GPT-5 被限制在相同的先验上下文中时，它可以至少与普通标注员一样好地再现人类的总体判断（附加模型在附录表 18 中报告）。

Corpus	Model	AIC (□)
INSQ	Base (word count)	472.58
Base+novelty		424.39
Base+relevance		444.29
Base+implication scope		445.92
Base+3 (all aspects)		416.60
Ablation(-implication scope)		416.01
Ablation(-relevance)		420.21
Ablation(-novelty)		438.26
FORA	Base (word count)	643.25
Base+novelty		599.28
Base+relevance		580.12
Base+implication scope		637.83
Base+3 (all aspects)		564.06
Ablation(-implication scope)		564.53
Ablation(-relevance)		601.09
Ablation(-novelty)		581.83

语料库	模型	AIC (□)
INSQ	基础（字数）	472.58
基础 + 新颖性		424.39
基础 + 相关性		444.29

基础 + 蕴含范围	445.92
基础 +3 (所有方面)	416.60
消融 (消融范围)	416.01
消融性 (相关性)	420.21
消融实验 (新颖性)	438.26
FORA	基础 (字数) 643.25
基础 + 新颖性	599.28
基础 + 相关性	580.12
基础 + 蕴含范围	637.83
基础 +3 (所有方面)	564.06
消融 (消融范围)	564.53
消融性 (相关性)	601.09
消融实验 (新颖性)	581.83

Table 4: Ordinal regression model comparison using AIC (lower is better). Base uses only utterance length. Base+3 includes all three aspects. Ablation models remove one aspect from the full model.

表 4：使用 AIC 进行序数回归模型比较（数值越小越好）。Base 仅使用语句长度。Base+3 包含所有三个方面。消融模型从完整模型中移除一个方面。

Corpus	Aspect	GPT-5	Human LOO
INSQ	CIG	0.457±0.020	0.656±0.131
Novelty	0.587±0.024	0.637±0.245	
Relevance	0.452±0.018	0.431±0.202	
Imp. Scope	0.529±0.021	0.562±0.162	
FORA	CIG	0.520±0.027	0.631±0.141
Novelty	0.556±0.017	0.599±0.200	
Relevance	0.414±0.015	0.446±0.163	
Imp. Scope	0.479±0.008	0.559±0.226	

语料库	方面	GPT-5	人类 LOO
INSQ	CIG	0.457±0.020	0.656±0.131
新颖性	0.587±0.024	0.637±0.245	
相关性	0.452±0.018	0.431±0.202	
重要范围	0.529±0.021	0.562±0.162	
FORA	CIG	0.520±0.027	0.631±0.141
新颖性	0.556±0.017	0.599±0.200	
相关性	0.414±0.015	0.446±0.163	
重要范围	0.479±0.008	0.559±0.226	

Table 5: Mean absolute error for GPT-5 when given only the memory summary—the same information provided to human annotators—compared to the Human LOO baseline.

表 5：当仅提供记忆摘要（即提供给人类标注员的相同信息）时，GPT-5 的平均绝对误差，与人类留一法基线相比。

We next examine how alternative prior-knowledge inputs reproduce the GPT-5_SUMMARY predictions, to check whether the memory-derived summary retains the information

我们接下来考察替代的先验知识输入如何再现 GPT-5_SUMMARY 的预测，以检查由记忆得出的摘要是否保留了信息

Corpus	Aspect	GPT-5_full	GPT-5_memory	GPT-5_short_prior	GPT-5_no_kn
INSQ	CIG	0.265±0.021	0.265±0.007	0.350±0.013	0.394±0.024
Novelty		0.353±0.009	0.321±0.028	0.649±0.038	0.764±0.044
Relevance		0.159±0.026	0.131±0.013	0.125±0.007	0.151±0.025
Imp. Scope		0.165±0.011	0.156±0.014	0.149±0.011	0.155±0.016
Mean		0.236±0.017	0.218±0.016	0.318±0.017	0.366±0.027
FORA	CIG	0.207±0.008	0.220±0.010	0.293±0.020	0.336±0.008
Novelty		0.279±0.006	0.259±0.019	0.383±0.031	0.492±0.025
Relevance		0.151±0.015	0.146±0.012	0.148±0.006	0.167±0.005
Imp. Scope		0.211±0.027	0.203±0.007	0.195±0.010	0.206±0.010
Mean		0.212±0.014	0.207±0.012	0.255±0.017	0.300±0.012
Overall mean (both corpora)		0.224±0.015	0.213±0.014	0.286±0.017	0.333±0.020

语料库	方面	GPT-5_full	GPT-5 记忆	GPT-5_short_prior	GPT-5_no_knowledge
INSQ	CIG	0.265±0.021	0.265±0.007	0.350±0.013	0.394±0.024
新颖性		0.353±0.009	0.321±0.028	0.649±0.038	0.764±0.044
相关性		0.159±0.026	0.131±0.013	0.125±0.007	0.151±0.025
重要范围		0.165±0.011	0.156±0.014	0.149±0.011	0.155±0.016
平均		0.236±0.017	0.218±0.016	0.318±0.017	0.366±0.027
FORA	CIG	0.207±0.008	0.220±0.010	0.293±0.020	0.336±0.008
新颖性		0.279±0.006	0.259±0.019	0.383±0.031	0.492±0.025
相关性		0.151±0.015	0.146±0.012	0.148±0.006	0.167±0.005
重要范围		0.211±0.027	0.203±0.007	0.195±0.010	0.206±0.010
平均值		0.212±0.014	0.207±0.012	0.255±0.017	0.300±0.012
总体均值（两个语料库）		0.224±0.015	0.213±0.014	0.286±0.017	0.333±0.020

Table 6: MAE of GPT-5 variants evaluated against GPT-5_summary reference, across both corpora and all aspects. Bold indicates the best performer across all variants.

表 6：在两个语料库和所有方面上，对 GPT-5 变体与 GPT-5_summary 参考进行评估的 MAE。粗体表示所有变体中表现最好的。

that most influences GPT-5’s ratings. Specifically, we compare four conditions: the full preceding transcript (GPT-5_full), the retrieved memory items shown verbatim without summarisation

(GPT-5_memory), only the three most recent utterances (GPT-5_short_prior), and no prior context beyond the topic (GPT-5_no_knowledge). As shown in Table 6, GPT-5_full and GPT-5_memory yield the smallest mean deviations from GPT-5_SUMMARY, while restricting or removing prior context produces substantially larger errors. Overall, this suggests that the memory-based summaries provide a compact yet faithful approximation of the information in the full history that drives GPT-5’s CIG estimates.

大多数影响 GPT-5 的评分。具体而言，我们比较了四种条件：完整的先前对话记录（GPT-5_full），以原文形式显示而没有摘要的检索记忆项（GPT-5_memory），仅有最近的三个发言（GPT-5_short_prior），以及不超出话题的先前上下文（GPT-5_no_knowledge）。如表 6 所示，GPT-5_full 和 GPT-5_memory 与 GPT-5_SUMMARY 的平均偏差最小，而限制或去除先前上下文则会产生显著更大的误差。总体而言，这表明基于记忆的摘要提供了对驱动 GPT-5 的 CIG 估计的完整历史中信息的紧凑而忠实的近似。

Comparing memory-dynamics signals to heuristic proxies We compare several informativeness-related heuristic proxies used in prior work—word-based, surprisal-based, entity-based, and TF-IDF features—against our memory-dynamics signals (per-utterance counts of extracted claims and claim updates) by computing Pearson correlations with the human-annotated CIG soft label (mean rating across annotators). As shown in Table 7, memory dynamics are the strongest correlates: relevance-gated updates (Memory changes (Relv⁺)) achieve the highest correlation ($|r| = 0.726$), followed by Memory changes (Any) ($|r| = 0.720$) and Extracted claim count ($|r| = 0.713$), outperforming all other proxies. Token surprisal sum and utterance Length (tokens) form a mid-tier baseline ($|r| = 0.678$ and $|r| = 0.697$, respectively), while entity-based signals are substantially weaker (e.g., Entity count $|r| = 0.466$). Because annotators were provided memory-derived prior context, memory dynamics may be advantaged; we therefore

将记忆动态信号与启发式代理进行比较我们将之前工作中使用的几种与信息量相关的启发式代理（基于词语的、基于惊讶度的、基于实体的以及 TF-IDF 特征）与我们的记忆动态信号（提取声明和声明更新的每句话计数）进行比较，通过计算与人工标注的 CIG 软标签（标注者平均评分）之间的皮尔逊相关性。如表 7 所示，记忆动态是最强的相关因素：基于相关性的更新（Memory changes (Relv⁺)) 取得了最高的相关性 ($|r| = 0.726$)，其次是 Memory changes (Any) ($|r| = 0.720$) 和 Extracted claim count ($|r| = 0.713$)，优于所有其他代理。令牌惊讶度总和和句子长度（令牌）构成了中等水平的基准（分别为 $|r| = 0.678$ 和 $|r| = 0.697$ ），而基于实体的信号则明显较弱（例如，Entity count $|r| = 0.466$ ）。由于标注者被提供了基于记忆的先前上下文，记忆动态可能具有优势；因此

Feature	$ r $ w/ CIG
Memory changes (Relv ⁺)	0.726
Memory changes (Any)	0.720
Extracted claim count	0.713
Memory changes (Info ⁺)	0.712
Memory changes (Novo ⁺)	0.703
TF-IDF sum	0.701
Length (tokens)	0.697
Token surprisal sum	0.678
TF-IDF max	0.638
Novel word count	0.601
Memory changes (Scope ⁺)	0.592
Entity count	0.466
Novel entity count	0.419
Novel entity ratio	0.298
Specificity (mean IDF)	0.223

功能	$\ r\ $ 与香烟一起使用
记忆变化 (Relv ⁺)	0.726
记忆变化 (任何)	0.720
提取的声明数量	0.713
记忆变化 (信息 ⁺)	0.712
记忆变化 (Novo ⁺)	0.703
TF-IDF sum	0.701
长度 (tokens)	0.697
Token 惊讶度总和	0.678
TF-IDF max	0.638
新词计数	0.601
记忆变化 (范围 ⁺)	0.592
实体数量	0.466
新实体数量	0.419
新实体比例	0.298
特异性 (平均 IDF)	0.223

Table 7: Pearson correlation between each proxy feature and the CIG average from human annotations. Aspect-gated memory-change features (marked with ⁺) count updates only for utterances whose corresponding aspect score is greater than 2.

表 7: 每个代理特征与人类注释的 CIG 平均值之间的皮尔逊相关性。方面门控记忆变化特征 (标记为 ⁺) 仅对相应方面得分大于 2 的句子进行计数更新。

replicate the same correlation analysis using GPT-5 predictions produced under full-transcript context (GPT-5_FULL) and observe a similar overall pattern (Appendix Table 19). Appendix B details the proxy definitions.

使用在完整转录本上下文中生成的 GPT-5 预测 (GPT-5_FULL) 重复进行相同的相关性分析, 并观察到相似的整体模式 (附录表 19)。附录 B 详细说明了代理变量的定义。

Claim-Level Predictions to Utterance Impression Motivated by broader interest in semantic salience (e.g., Biggs et al., 2012), we conduct an exploratory analysis to examine how claim-level qualities roll up to utterance-level impressions of CIG (Wei et al., 2022). First, we use GPT-5 to predict scores (1–4) for every extracted claim on all three aspects (Novelty, Relevance, and Scope; see Appendix Table 17 for prompt details). We then apply a two-step, unsupervised procedure to test

基于对语义显著性更广泛兴趣 (例如, Biggs 等, 2012), 我们进行了一项探索性分析, 以考察声明层面的质量如何汇总为对 CIG 的语句层面印象 (Wei 等, 2022)。首先, 我们使用 GPT-5 为所有三个方面的每个提取声明预测分数 (1–4) (新颖性、相关性和范围; 详见附录表 17 中的提示细节)。然后, 我们应用一个两步的无监督过程来测试

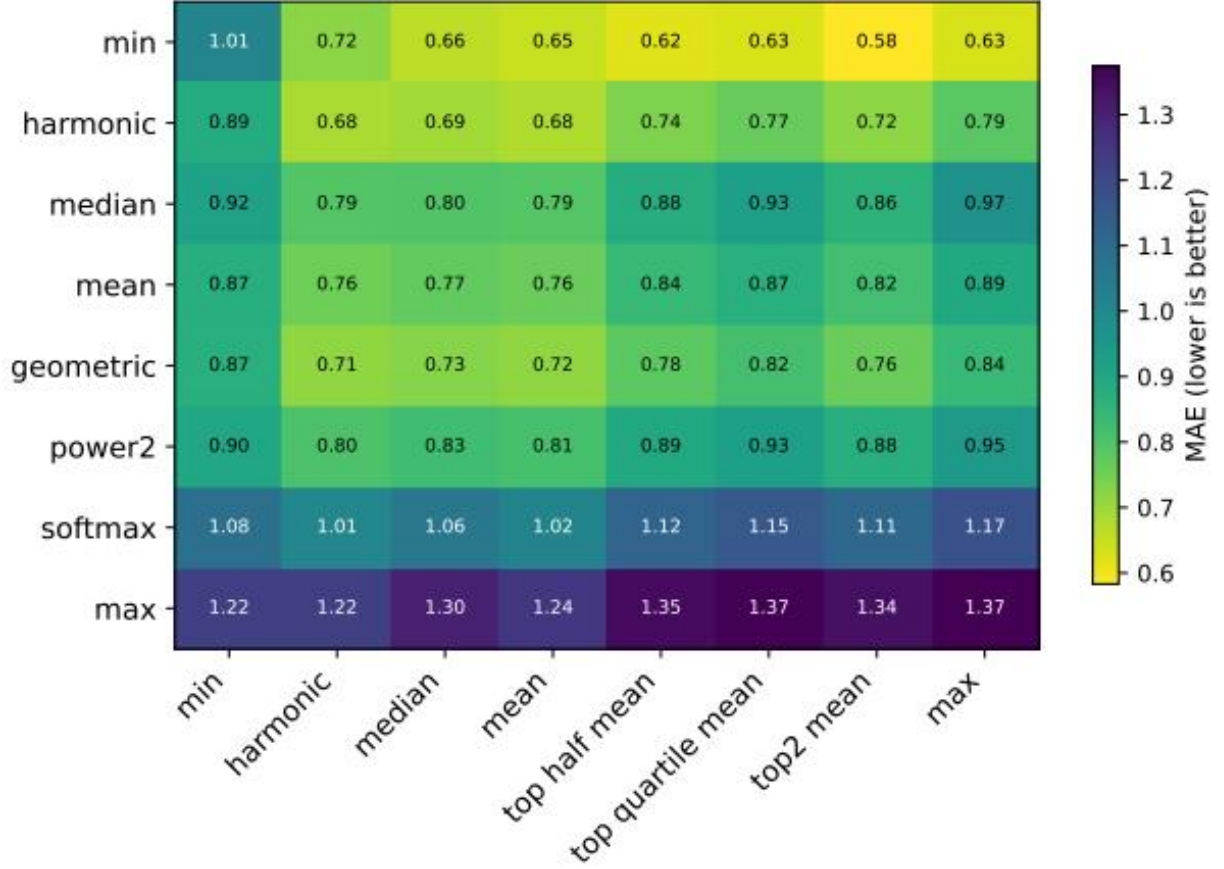


Figure 2: images/d70cb05d5cbea33588a1767182947fe339fce5bdc76ec80cfad3c22d533e1f93.jpg

Figure 2: Heatmap of MAE for predicting human utterance-level CIG. The y-axis represents Aspect Combination methods. The x-axis represents Claim Aggregation methods.

图 2：预测人类语句级 CIG 的 MAE 热图。y 轴表示方面组合方法。x 轴表示主张聚合方法。

how well these claim-level predictions can recover human-annotated utterance-level CIG scores. This involves: (i) aggregating aspect-specific scores across multiple claims within an utterance (claim-aggregation on the x-axis), and (ii) combining the resulting aspect scores into a single CIG estimate (aspect-combination on the y-axis). We test a range of operators for both steps.

这些声明级预测在多大程度上能够恢复人工标注的句子级 CIG 评分。这涉及以下两个步骤：(i) 在一个句子内，对多个声明中的方面特定评分进行聚合（x 轴上的声明聚合），以及 (ii) 将得到的方面评分组合成一个单一的 CIG 估计值（y 轴上的方面组合）。我们对这两个步骤都测试了一系列运算符。

The results, shown in Figure 2, reveal two consistent patterns. First, for aspect combination (y-axis), the “min” operator performs best by a large margin across all claim-level aggregators. This suggests a conjunctive bottleneck: utterance-level informativeness is effectively constrained by the weakest aspect, rather than increasing smoothly with the average strength of all aspects. Second, for claim aggregation (x-axis), the best-performing operators are selective pooling methods such as “top-2 mean” (which achieves the lowest MAE of 0.583) and “top-quartile mean” (MAE = 0.627), especially when paired with “min”. By contrast, combination operators that are more sensitive to high values, such as “softmax” and “max”, perform worst. Taken together, these findings suggest that listeners do not judge an utterance by uniformly averaging all of its claims. Instead, impressions of CIG appear to be driven by a small number of salient claims, while still requiring that no core aspect be critically weak. This supports our broader view that perceived informativeness in dialogue is multi-dimensional and bottlenecked, rather than a simple additive function

of semantic content.

如图 2 所示的结果揭示了两个一致的模式。首先，在方面组合（y 轴）方面，“min”操作符在所有主张级聚合器中表现最佳，优势非常明显。这表明存在一个连词瓶颈：话语层面的信息量实际上受到最弱方面的限制，而不是随着所有方面平均强度的增加而平滑增长。其次，在主张聚合（x 轴）方面，表现最好的操作符是选择性池化方法，例如“top-2 平均”（实现了最低的 MAE 为 0.583）和“top-quartile 平均”（MAE = 0.627），尤其是在与“min”结合使用时。相比之下，对高值更敏感的组合操作符，例如“softmax”和“max”，表现最差。总体来看，这些发现表明，听众并不通过统一平均所有主张来判断话语。相反，CIG 的印象似乎受到少量显著主张的驱动，同时仍然要求没有任何核心方面过于薄弱。这支持了我们更广泛的观点，即对话中感知的信息量是多维且存在瓶颈的，而不是语义内容的简单相加函数。

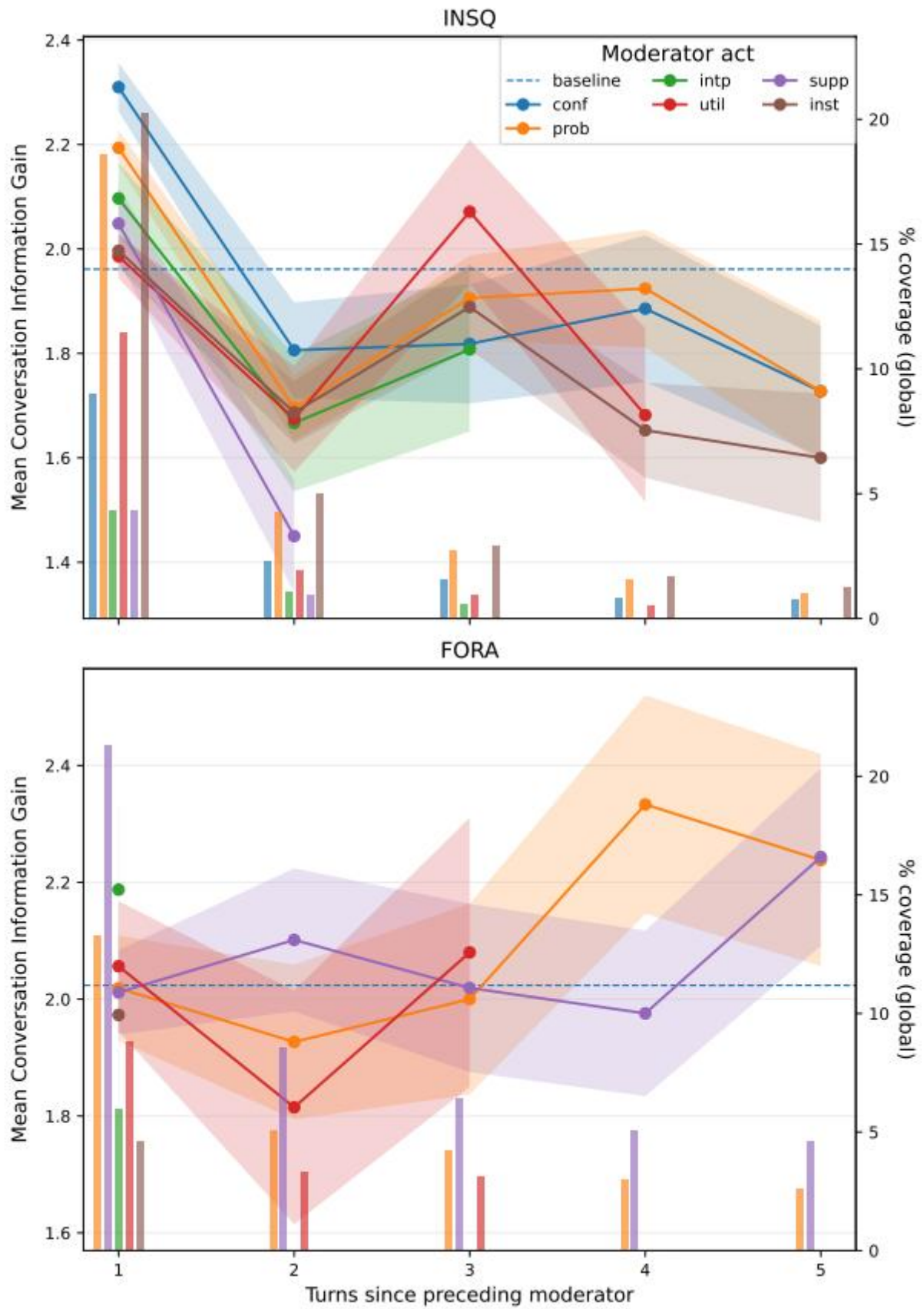


Figure 3: Mean participant CIG (left y-axis) vs. turns since the last moderator act (x-axis), for both settings. Lines show the mean CIG for each moderator act. Bars (right y-axis) show the global % coverage for each act. The dashed line is the corpus mean CIG.

图 3: 平均参与者 CIG (左 y 轴) 与最后一次主持人行为后的回合数 (x 轴) 的关系, 适用于两种设置。线条显示了每次主持人行为的平均 CIG。条形图 (右 y 轴) 显示了每次行为的全局 % 覆盖率。虚线是语料库平均 CIG。

8 Case Study: Moderator Dynamics

8 案例研究：主持人动态

To demonstrate CIG’s utility beyond static evaluation, we examine whether it can diagnose interaction dynamics—specifically, how moderator interventions shape subsequent informational progress. Facilitators play a central role in deliberation, yet their effects are usually described qualitatively or through broad outcome measures rather than through the local informational gain they elicit. To categorize these interventions, we employ the WHoW taxonomy (Chen et al., 2024), a domain-agnostic framework designed to analyze moderator behavior across varying contexts. WHoW classifies moderator turns into six broad communicative acts: Probing (eliciting responses), Confronting (direct engagement between participants), Interpretation (interpreting previous content), Supplementing (contributing information), and Utilities (other functional acts). By mapping downstream CIG trajectories to these specific acts, we can isolate which strategies trigger informational gain and how these effects vary by context.

为了展示 CIG 在静态评估之外的用途, 我们考察它是否能够诊断互动动态——具体而言, 调节者干预如何塑造后续的信息进展。促进者在讨论中扮演着核心角色, 但其影响通常仅通过定性描述或广泛的结果指标来描述, 而非通过它们所引发的局部信息增益。为了对这些干预进行分类, 我们采用 WHoW 分类法 (Chen 等, 2024), 这是一个与领域无关的框架, 旨在分析调节者行为在不同情境下的表现。WHoW 将调节者的发言分为六种广泛的交流行为: 探询 (引发回应)、对抗 (参与者之间的直接互动)、解释 (解释先前内容)、补充 (提供信息) 以及实用功能行为 (其他功能性行为)。通过将后续的 CIG 轨迹映射到这些具体行为, 我们可以确定哪些策略会引发信息增益, 以及这些效果如何因情境而异。

To quantify these dynamics, we align each participant utterance to the most recent moderator turn. These pairs are grouped by moderator act type and indexed by their temporal lag—specifically, the number of turns (1–5) following the intervention. We then compute the mean participant CIG at each lag for every act type while accounting for the overall prevalence of each act within the corpus.

为了量化这些动态, 我们将每位参与者的话语与最近一次主持人的发言进行对齐。这些对子按照主持人的行为类型进行分组, 并以其时间滞后作为索引——具体来说, 即干预后跟随的发言次数 (1–5 次)。然后, 在考虑语料库中每种行为的整体出现频率的情况下, 我们计算每种行为类型在每个滞后下的平均参与者 CIG 值。

As shown in Figure 3, a clear divergence emerges between the two settings. In INSQ, moderator interventions are followed by sharp, immediate CIG peaks, with the strongest gains typically occurring at lag 1. This is particularly evident in Confronting acts, which align with the adversarial debate format where moderators actively sharpen disagreement to prompt direct rebuttals. In FORA, by contrast, peaks are more diffused and often emerge several turns later. This suggests that in community discussions, informational progress is less tethered to a single facilitator move, instead accumulating gradually through a sequence of follow-on contributions. Notably, Confronting acts are absent in FORA, consistent with its less adversarial and more facilitative interaction style.

如图 3 所示, 两种设置之间出现了明显的差异。在 INSQ 中, 调节者干预后会出现明显的、即时的 CIG 峰值, 通常在滞后 1 时达到最强。这在“对抗性行为”中尤为明显, 这种行为符合对抗性辩论的格式, 其中调节者积极加剧分歧以促使直接反驳。相比之下, 在 FORA 中, 峰值更加分散, 并且通常在多个回合之后才出现。这表明在社区讨论中, 信息性进展较少依赖于单一调节者的行动, 而是通过一系列后续贡献逐渐累积起来。值得注意的是, FORA 中不存在“对抗性行为”, 这与其较少对抗性、更具引导性的互动风格一致。

This contrast shows that CIG is sensitive not only to the content of individual utterances but also to the interactional regime in which they are produced. In INSQ, moderators manage turn-taking

and actively challenge participants within a competitive structure, so interventions often elicit immediate high-information responses. In FORA, facilitation is looser and more participatory, with more Supplementing moves from facilitators themselves, so informational progress appears more distributed across subsequent turns. More broadly, this case study illustrates how CIG can serve as a diagnostic lens for comparing moderation styles, revealing not only whether a discussion is informative overall, but also how information gain is locally elicited and sustained over time.

这种对比表明，CIG 不仅对单个话语的内容敏感，还对产生这些话语的互动机制敏感。在 INSQ 中，主持人管理发言顺序，并在一个竞争性的结构中积极挑战参与者，因此干预常常会引发立即的高信息量回应。在 FORA 中，引导更为宽松和参与性更强，主持人自身更多的补充性发言，因此信息进展似乎在后续的发言中更为分散。更广泛地说，这个案例研究说明了 CIG 如何可以作为一种诊断工具，用于比较不同的主持风格，不仅揭示讨论整体是否具有信息性，还说明信息获取是如何在局部层面被引发并持续一段时间的。

9 Conclusions

9 结论

We introduced CIG, a framework for measuring Conversational Information Gain in deliberative dialogue by focusing on informational progress rather than surface form. Grounded in the notion of information gain as a change in a shared knowledge state (Meadow and Yuan, 1997), our approach operationalizes CIG through a lightweight semantic memory maintained via LLM-based claim extraction and NLI-guided consolidation. This evolving record of claims provides the necessary context to evaluate utterances along four interpretable dimensions: overall CIG, Novelty, Relevance, and Implication Scope.

我们引入了 CIG（对话中信息增益的衡量框架），该框架通过关注信息进展而非表层形式来衡量协商对话中的信息增益。我们的方法基于信息增益作为共享知识状态变化的概念（Meadow 和 Yuan, 1997），通过基于大型语言模型（LLM）的主张提取和基于自然逻辑推理（NLI）引导的整合，实现对 CIG 的轻量级语义记忆。这一不断演进的主张记录为评估话语提供了必要的上下文，评估维度包括四个可解释的维度：总体 CIG、新颖性、相关性和影响范围。

Across two contrasting deliberative settings—INSQ debates and FORA community discussions—we found that CIG can be annotated with moderate-to-high reliability and meaningfully decomposed into interpretable aspects. Novelty and Relevance emerged as the primary drivers of perceived CIG, while Implication Scope was more informative as a descriptive dimension of deliberative style than as a consistent predictor of informativeness. GPT-5 was able to reproduce aggregate human judgments under matched information conditions, suggesting that CIG can be assessed automatically with reasonable fidelity.

在两种对比的协商性情境——INSQ 辩论和 FORA 社区讨论中，我们发现 CIG 可以以中等到较高的可靠性进行标注，并且可以有意义地分解为可解释的方面。新颖性和相关性被发现是感知到的 CIG 的主要驱动因素，而影响范围作为协商风格的描述性维度比作为信息性的一致预测因素更具信息量。在信息条件匹配的情况下，GPT-5 能够再现人类的总体判断，这表明 CIG 可以以合理的准确性自动进行评估。

From a broader perspective, these results highlight the value of tracking semantic memory dynamics directly. Memory-based signals—such as counts of extracted claims and claim updates—aligned more strongly with human-perceived CIG than common heuristic proxies. Our analysis further suggests that informativeness perception is governed by a conjunctive bottleneck: an utterance is only perceived as informative as its weakest dimension allows. Finally, the moderation case study demonstrates CIG’s utility as a diagnostic lens. In INSQ, moderator interventions catalyzed immediate informational surges, whereas in FORA, gains accumulated more gradually across subsequent turns. Together, these findings position CIG as a framework for studying how information is introduced, elicited, and sustained in deliberative interaction.

从更广泛的角度来看，这些结果突显了直接追踪语义记忆动态的价值。基于记忆的信号——例如提取声明的数量和声明更新——与人类感知的 CIG（信息获取感）相比，与常见的启发式代理指标更为一致。我们

的分析进一步表明, 信息丰富性感知受到一个联合瓶颈的制约: 一个陈述仅被认为是具有信息量的, 取决于其最弱的维度。最后, 调节案例研究展示了 CIG 作为诊断透镜的实用性。在 INSQ 中, 调节干预引发了即时的信息激增, 而在 FORA 中, 信息收益则在后续回合中逐渐累积。这些发现共同将 CIG 定位为研究信息如何在协商性互动中被引入、引发和维持的框架。

Future work could integrate newer memory architectures (Xu et al., 2025; Li et al., 2025) to improve consolidation robustness and long-context tracking. The NLI-based update mechanism may also support analyses of finer-grained meaning construction in multi-party dialogue (Poole-Dayanan et al., 2025). Furthermore, memory-based information gain signals may provide interpretable supervision or even reward signals for training and evaluating information-seeking, deliberation-support, and facilitation-oriented conversational systems.

未来的工作可以整合更新的内存架构 (Xu 等, 2025; Li 等, 2025), 以提高巩固的鲁棒性和长上下文跟踪能力。基于自然语言推理 (NLI) 的更新机制还可能支持对多方对话中更细粒度意义构建的分析 (Poole-Dayanan 等, 2025)。此外, 基于内存的信息增益信号可能为训练和评估以信息获取、推理支持和促进为导向的对话系统提供可解释的监督甚至奖励信号。

Limitations

限制

Our study has several limitations. First, the CIG labels are collected under a constrained information condition (topic, a prior-knowledge summary,

我们的研究有几个局限性。首先, CIG 标签是在受限信息条件下收集的 (主题, 先验知识摘要,

and a short preceding window), which may bias judgments toward what is captured by the summary rather than the full discourse; while we partially probe this with context ablations, the results may not generalize to settings where annotators have full access to long histories. Second, the semantic-memory pipeline relies on LLM extraction and NLI-style consolidation, so errors in claim parsing, retrieval, or entailment can propagate into memory dynamics and any downstream analyses. Third, due to efficiency constraints, each extracted claim is assigned only a single memory action, and at most one relation to one existing memory item is recorded, which can under-represent cases where a claim simultaneously relates to multiple prior claims. Finally, we study two moderated English-language corpora and exclude highly sensitive topics; broader coverage (languages, cultures, and adversarial/polarized domains) is needed to assess robustness and external validity.

以及一个较短的前序窗口), 这可能会使判断偏向于摘要所捕捉的内容, 而不是完整的论述; 虽然我们通过上下文消去实验部分地探讨了这一点, 但结果可能无法推广到标注者能够完整访问长期历史的场景。第二, 语义记忆管道依赖于 LLM 提取和 NLI 风格的整合, 因此在主张解析、检索或蕴含过程中出现的错误可能会传播到记忆动态以及任何后续分析中。第三, 由于效率限制, 每个提取的主张仅被分配一个单一的记忆动作, 并且最多记录与一个现有记忆项的一个关系, 这可能会低估那些主张同时与多个先前主张相关的情况。最后, 我们研究了两个经过调节的英语语料库, 并排除了高度敏感的话题; 为了评估稳健性和外部效度, 需要更广泛的覆盖范围 (包括语言、文化以及对抗性/极化领域)。

Ethical considerations

伦理考量

We recruited annotators via an online platform to rate the informativeness of dialogue segments drawn from public deliberative settings. Participation was voluntary and based on informed consent; annotators were informed of their right to withdraw at any time without penalty and could opt out of rating specific segments after viewing the topic if they felt uncomfortable (Appendix D show anonymised version of the plain language statement for participation). Because public dis-

cussions can include sensitive or emotional content, we limited topic selection and provided sufficient context to support informed skipping. We collected no direct personal identifiers beyond platform-managed participant IDs, and we release only de-identified annotations. Compensation followed platform guidelines and was set to meet or exceed applicable minimum-wage expectations for estimated task duration. The annotation protocol and material were approved by the University of Melbourne research ethics committee with the reference code- 2023-28400-47354-1. We also acknowledged that the studies have used large language model, including Gemini-2.5-pro from Google and GPT-5 from OpenAI, for code refactoring and data cleaning.

我们通过一个在线平台招募标注员，以评估从公共协商场景中提取的对话片段的信息量。参与是自愿的，并基于知情同意；标注员被告知他们随时可以无惩罚地退出，并且在观看主题后如果感到不适，可以选择不对特定片段进行评分（附录 D 展示了参与的通俗语言声明的匿名版本）。由于公共讨论可能包含敏感或情感内容，我们限制了主题选择，并提供了足够的上下文以支持知情跳过。我们收集的个人直接标识符仅限于平台管理的参与者 ID，我们仅发布去标识化的标注结果。报酬遵循平台指南，并设定为达到或超过估计任务时长适用的最低工资标准。该标注协议和材料已获得墨尔本大学研究伦理委员会的批准，参考代码为-2023-28400-47354-1。我们还承认研究中使用了大型语言模型，包括谷歌的 Gemini-2.5-pro 和 OpenAI 的 GPT-5，用于代码重构和数据清理。

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